



BENGALURU NORTH UNIVERSITY
TAMAKA, KOLAR - 563103

Curriculum/Syllabus
for
Postgraduate Programme

**Master of
COMPUTER APPLICATIONS
(MCA)**

Choice Based Credit System

Faculty of Science

(With Effect from Academic Year 2024–2025)



BENGALURU NORTH UNIVERSITY

Sri Devaraj Urs Extension, Tamaka, Kolar – 563 103.

No: BNU /BOS/P.G- Computer Science / 2024-25

Date: 13.05.2024

NOTIFICATION

Sub: Constitution of Board of Studies in **Computer Science /MCA (PG)** – reg

Ref: 1. Resolution of the Syndicate meeting held on: 06.02.2024

2. Approval of the Vice-Chancellor Dated: 01.05.2024

Pursuant to the resolution of the Syndicate cited at reference (1) and Under Section 33 of the Karnataka State Universities Act 2000 and Bangalore University Statutes, the Board of Studies in **Computer Science /MCA (PG)** for Bengaluru North University is constituted as follows with immediate effect for a period of 3 years or until further orders.

1.	Dr. Hanumanthappa Professor of Computer Science, Bangalore University.	Chairman	9448819017
2.	Dr. Murugan Dept. of Computer Science , Government First Grade College, KR .Puram, Bengaluru.	Member	9449880799
3.	Prof. Hidyathulla Dept. of Computer Applications Indo Asian Academy Degree College, Bengaluru.	Member	7406880932
4.	Smt. Vijay Mala Dept. of Computer Applications Krupanidhi Degree College, Chikkabellandur Bengaluru	Member	9886603147
5.	Dr. Manjunath Dept. of Computer Science, S.E.A College of Management & Science Bengaluru	Member	9844864295
6.	Mr. Paul Dinesh Senior Software Engineer Infosys Bengaluru	Member	9972345790
7.	Dr. Deepa T Nagalavi Dept. of Computer Science St. Joseph University of Bengaluru. Mail Id: deeoanagalavi@gmail.com	Member	9880631087
8.	Smt. Hamsalatha J Faculty of Computer Science PG Centre Mangasandra, Bengaluru North University	Co-ordinate	9481587447

As per the decision of the Syndicate, the Hon'ble Vice-Chancellor is authorized to nominate two experts from the Industry/NGOs/State/National Organizations.



The Chairman and Members are requested to accept the Chairmanship/ Membership of the above Board.

The Sitting fee, local conveyance allowance and TA/DA as the case may be, shall be paid to the Chairperson and Members of the Board as per the Bengaluru North University Rules for attending the BOS meetings.


REGISTRAR
Registrar

Bengaluru North University

To:

The Chairperson and Members of the Board of Studies in Computer Science and Information Technology (PG) BNU
Tamaka, Kolar, Karnataka 563103.

Copy to:

1. PS to Vice-Chancellor, Bengaluru North University, Tamaka, Kolar 563103
2. Registrar (Evaluation), Bengaluru North University, Tamaka, Kolar 563103
3. The Finance Officer, Bengaluru North University, Tamaka, Kolar 563103
4. Office copy/Guard file

REGULATIONS OF MASTER OF COMPUTER APPLICATIONS (MCA) COURSE

1. TITLE OF THE COURSE: Master of Computer Applications (MCA)

2. DURATION OF THE COURSE: The course of study shall be two years (4 semesters).

3. ELIGIBILITY FOR ADMISSION: A candidate with any degree of a minimum of 3 years duration (10+2+3) of Bangalore university or of any other University equivalent there in to with a minimum of 50% of marks in the aggregate of all subjects including languages, if any, provided further, that the candidate has studied Mathematics / Computer science /Business Mathematics / Statistics / Computer Applications / Electronics as a subject at PUC level or equivalent HSC (XII Standard) or at Degree level is eligible for admission to MCA Course. Relaxation to SC/ST, Group I be extended as per University norms.

4. ATTENDANCE: In each Semester a candidate should be considered to have successfully undergone the prescribed Course of study if the candidate has attended at least 75% of the classes in each subject (Theory, Lab & Practical).

5. MEDIUM OF INSTRUCTION: English.

6. SCHEME OF EXAMINATION:

A. The Internal Assessment marks should be decided for each of the theory subjects by conducting 2 tests, each of 60 minutes duration, spread over the span of a Semester. A seminar should also be given by the student in the second year and the same to be assessed and evaluated for internal assessment along with two tests.

B. The Internal Assessment marks in Practical course is based on the performance in the Laboratory. The Internal Assessment marks for Project work of a candidate is based on the dissertation and seminar.

7. ELIGIBILITY TO GO TO THE HIGHER SEMESTER:

A. A Candidate is allowed to carry over all the previous uncleared (failed) theory papers and Practicals to subsequent semesters from the first to fourth semester.

B. The maximum period for the completion of the course shall be four years from the date of admission.

8. MINIMUM FOR PASS AND DECLARATION OF RESULTS

A. For a pass in a semester, a candidate shall secure a minimum of 40% of the marks prescribed for a subject in the University Examination (Theory, Practical, Project work) and 50% of the marks in the aggregate inclusive of the Internal Assessment marks obtained in all subjects put together.

B. The candidates who do not satisfy 7(A) shall be deemed to have failed and have to take exams in the subjects in which he has secured less than 40% at the University examination.

C. Provision is made for rejection of results of all the subjects of a Semester only once, if the candidate decides to reappear for all the subjects of that semester. Such rejection should be made within 30 days of announcement of result, by making a written application, through the Head of the Institution. If such rejection is in respect of the results of all the subjects of one semester and earn fresh Internal marks as well.

D. The results of any semester will be declared as pass or fail as the case may be in accordance with regulation 7(A).

E. To be eligible for the award of the MCA degree, a candidate shall have completed the scheme of training and passed in all subjects prescribed for the Course.

F. Further to regulation 7(A), the classification followed by the University for all PG courses shall be made applicable for the declaration of results of each Semester.

9. CLASSIFICATION OF RESULT FOR THE MCA COURSE AND DECLARATION OF RANKS: Further to regulations 7(A) and 7(F), the names of all successful candidates securing First Class with Distinction and First Class in the First attempt shall be arranged in the order of Merit and only first FIVE Ranks shall be declared.

Any other issue not envisaged above shall be resolved by the Vice-Chancellor in consultation with the appropriate bodies of the University, which shall be final and binding.

10. QUESTION PAPER PATTERN:

The question paper pattern for theory paper has two sections. (70 Marks)

1. Section A includes 8 questions, students has to attend 5 questions. Each carries 6 Marks ($5 * 6 = 30$)
2. Section B includes 6 questions (question may contain sub questions), students has to attend 4 questions. each carries 10 Marks ($4 * 10 = 40$)

11. INTERNAL ASSESSMENT SCHEME (THEORY AND PRACTICAL)

Assessment Criteria	Marks
Attendance	10
Test	10
Assignment	5
Seminar	5
Total	30

**SCHEME OF STUDY AND EXAMINATION FOR MASTER OF COMPUTER APPLICATIONS
(MCA)**

Semester	Paper Code	Title of the Paper	Hours /Week	Marks			Credits	
				IA	Exam	Total	Subject	Total
I	MCA101T	Object Oriented Programming with Java	4	30	70	100	4	28
	MCA102T	Advanced Software Engineering	4	30	70	100	4	
	MCA103T	Mathematical Foundations	4	30	70	100	4	
	MCA104T	Advanced Database Management System	4	30	70	100	4	
	MCA105T	Data Structures and Algorithms	4	30	70	100	4	
	MCA106T	Theory of Computation	4	30	70	100	4	
	MCA107P	Java Programming Lab	4	30	70	100	2	
	MCA108P	Data Structures and Algorithm Lab	4	30	70	100	2	
II	MCA201T	Artificial Intelligence	4	30	70	100	4	28
	MCA202T	Web Technologies	4	30	70	100	4	
	MCA203T	Advanced Python Programming	4	30	70	100	4	
	MCA204T	Operating System and Linux	4	30	70	100	4	
	MCA205T	Network and Information Security	4	30	70	100	4	
	MCA206T	Cloud Computing	4	30	70	100	4	
	MCA207P	Web Technologies Lab	4	30	70	100	2	
	MCA208P	Python Programming Lab	4	30	70	100	2	
III	MCA301T	Research Methodology	4	30	70	100	4	28
	MCA302T	Data Science	4	30	70	100	4	
	MCA303T	Elective-I	4	30	70	100	4	
	MCA304T	Elective-II	4	30	70	100	4	
	MCA305T	Elective-III	4	30	70	100	4	
	MCA306P	Data Science Lab	8	30	70	100	4	
	MCA307P	Mini Project	8	30	70	100	4	
IV	MCA401P	Main Project	32	120	280	400	16	16

Electives

Elective-I (Any one to be opted)	Machine Learning
	Object Oriented Analysis and Design using UML
	Internet of Things
Elective-II (Any one to be opted)	Cyber Security and Cyber Laws
	Software Testing Tools
	Mobile Computing
Elective-III (Any one to be opted)	Big Data Analytics
	Deep Learning
	Blockchain Technologies

Program Outcome:

Sl. No.	Description	POs
1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and computer science and business systems to the solution of complex engineering and societal problems.	PO1
2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering and business problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	PO2
3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	PO3
4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	PO4
5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations	PO5
6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering and business practices.	PO6
7	Environment and sustainability: Understand the impact of the professional engineering solutions in business societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	PO7
8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering and business practices.	PO8
9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	PO9
10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	PO10
11	Project management and finance: Demonstrate knowledge and understanding of the engineering, business and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	PO11
12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	PO12

FIRST SEMESTER

MCA101T: Object Oriented Programming with Java

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Understand the basic object oriented programming concepts and apply them in problem solving.
- Use object oriented programming concepts to solve real world problems.
- Explain the concept of class and objects with access control to represent real world entities.
- Demonstrate the behavior of programs involving the basic programming constructs like control structures, constructors, string handling and garbage collection.
- Use overloading methodology on methods and constructors to develop application programs.
- Demonstrate the implementation of inheritance (multilevel, hierarchical and multiple) by using extend and implement keywords.
- Describe the concept of interface and abstract classes to define generic classes.
- Use dynamic and static polymorphism to process objects depending on their class Understand the basics of java console and GUI based programming

Content	Hours
Unit 1: Introduction Evolution of programming methodologies, Procedural Approach vs Object-Oriented Approach. Principles of OOP - Encapsulation, Inheritance and Polymorphism. Concepts of OOP - Abstraction, Overloading, Reusability, Extensibility, Dynamic Binding, Message Passing. Java Features; comparison of Java with C and C++; Java and Internet; Java Environment; Java Program structure; Java Tokens; Implementing a Java Program Java Virtual Machine; Constants, Variables; Data Types; Scope of Variables; Type casting; Operators and expressions; Decision Making; Branching and Looping.	12
Unit 2: Classes and Arrays Defining a class; Constructors; Methods; overloading; static Members; Nesting of Methods; Overriding methods; Final Classes; Abstract Class; Visibility control; Arrays; creating an array; Two Dimensional arrays; Strings; String Arrays; String Methods; StringBuffer Class; Vectors; Wrapper Classes.	10
Unit 3: Inheritance Interfaces and Packages Defining a subclass; Subclass constructor; Types of Inheritance, multilevel inheritance; Hierarchical Inheritance; Defining Interfaces; Extending Interfaces; Implementing Interfaces; Java API Packages; creating a package; Accessing and using a package; adding class to a package; Hiding Classes.	10
Unit 4: Multithreading Exception Handling and Files Creating Threads Extending the Thread class; Thread Life cycle; Thread Exception, Thread priority, Synchronization; Runnable Interface; Exceptions, Throwing own Exceptions; Concepts of streams; stream classes; Byte Stream Classes; Character stream Classes; Using Streams; Using file Class; Other Stream Classes.	10
Unit 5: Swing, JDBC Introduction to Swing, Swing Controls, Adapter Class, Event handling mechanism, Java Database Connectivity: introduction; Establishing Connection, Creation of Data Tables; Table Updating, Result set.	10

References Books:

1. Herbert Schildt and Dale Skrien, "Java Fundamentals – A comprehensive Introduction", McGraw Hill, 1st Edition, 2013.
2. Herbert Schildt, "Java the complete reference", McGraw Hill, Osborne, 7th Edition, 2011.
3. T.Budd, "Understanding Object- Oriented Programming with Java", Pearson Education, Updated Edition (New Java 2 Coverage), 1999.
4. P.J.Dietel and H.M.Dietel, "Java How to program", Prentice Hall, 6th Edition, 2005.
5. P.Radha Krishna, "Object Oriented programming through Java", CRC Press, 1st Edition, 2007.
6. S.Malhotra and S. Choudhary, "Programming in Java", Oxford University Press, 2nd Edition, 2014

Course outcome (Course Skill Set)

At the end of the course the student will be able to:

Sl. No.	Description	Blooms Level
CO1	Identify, analyze and define database objects, enforce integrity constraints on a database using RDBMS	L2
CO2	Use Structured Query Language (SQL) for database manipulation and also demonstrate the basic of query evaluation.	L2
CO3	Design and build simple database systems and relate the concept of transaction, concurrency control and recovery in database	L3
CO4	Develop application to interact with databases, relational algebra expression.	L3
CO5	Develop applications using tuple and domain relation expression from queries.	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	x		x									
CO2		x			x							
CO3	x		x									
CO4	x			x								
CO5		x	x									

MCA102T: Advanced Software Engineering

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Outline software engineering principles and activities involved in building large software programs.
- Identify ethical and professional issues and explain why they are of concern to software engineers.
- Explain the fundamentals of object oriented concepts.
- Describe the process of requirements gathering, requirements classification, requirements specification and requirements validation.
- Differentiate system models, use UML diagrams and apply design patterns.
- Discuss the distinctions between validation testing and defect testing.

Content	Hours
Unit 1: Basics of Software Testing and Examples Basic definitions, Test cases, Insights from a Venn diagram, Identifying test cases, Error and fault taxonomies, Levels of testing. Examples: Generalized pseudocode, The triangle problem, The Next Date function, The commission problem, The SATM (Simple Automatic Teller Machine) problem. Decision Table-Based Testing: Decision tables, Test cases for the triangle problem, Test cases for the Next Date function, Test cases for the commission problem, Guidelines and observations. Data Flow Testing: Definition-Use testing, Slice-based testing, Guidelines and observations.	12
Unit 2: Levels of Testing Levels of Testing: Traditional view of testing levels, Alternative life-cycle models, The SATM system, Separating integration and system testing. Integration Testing: A closer look at the SATM system, Decomposition-based, call graph-based, Path-based integrations, Case study. System Testing: Threads, Basic concepts for requirements specification, Finding threads, Structural strategies and functional strategies for thread testing, SATM test threads, System testing guidelines, ASF (Atomic System Functions) testing example.	10
Unit 3: Interaction Testing Context of interaction, taxonomy of interactions, Interaction, composition, and	10

determinism, Client/Server Testing. Issues in Object-Oriented Testing: Units for object-oriented testing, Implications of composition and encapsulation, inheritance, and polymorphism, Levels of object-oriented testing, GUI testing, Dataflow testing for object-oriented software, Examples. Class Testing: Methods as units, Classes as units.	
Unit 4: Object-Oriented Integration Testing UML support for integration testing, MM-paths for object-oriented software, A framework for object-oriented dataflow integration testing. GUI Testing: The currency conversion program, Unit testing, Integration Testing and System testing for the currency conversion program. Object-Oriented System Testing: Currency converter UML description, UML-based system testing, Statechart-based system testing. Exploratory Testing: The context-driven school, Exploring exploratory testing, Exploring a familiar example, Exploratory and context-driven testing observations.	10
Unit 5: Model-Based Testing Testing based on models, Appropriate models, Use case-based testing, Commercial tool support for model-based testing. Test-Driven Development: Test-then-code cycles, Automated test execution, Java and JUnit example, Remaining questions, Pros, cons, and open questions of TDD, Retrospective on MDD versus TDD. A Closer Look at All Pairs Testing: The all-pairs technique, A closer look at NIST study, Appropriate applications for all pairs testing, Recommendations for all pairs testing. Software Testing Excellence: Craftsmanship, Best practice of software testing.	10

Reference Books:

1. Roger S. Pressman – Software Engineering, A Practitioner’s approach, 7th Edition, McGRAW-HILL Publication, 2010.
2. Pankaj Jalote, “An integrated approach to Software Engineering”, 3rd Edition, Narosa Publishing House, 2013.
3. Ian Sommerville – Software Engineering, 9th Edition, Pearson Education Ltd, 2010.
4. Stephan R. Schach, “Object oriented software engineering”, Tata McGrawHill, 2008

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Design a software system, component or process to meet desired needs within realistic constraints	L2
CO2	Assess professional and ethical responsibility	L1
CO3	Function on multi-disciplinary teams	L1
CO4	Use the techniques, skills, and modern engineering tools necessary for engineering practice	L2
CO5	Analyze, design, implement, verify, validate, implement, apply, and maintain software systems or parts of software systems	L4

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1			X									
CO2								X				
CO3			X							X		
CO4					X							
CO5	X											

MCA103T: Mathematical Foundations

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- To introduce the concepts of mathematical logic.
- To introduce the concepts of sets, relations, and functions.
- To perform the operations associated with sets, functions, and relations.
- To relate practical examples to the appropriate set, function, or relation model, and interpret the associated operations and terminology in context.
- To use Graph Theory for solving problems.

MCA104T: Advanced Database Management System

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Provide a strong foundation in database concepts, technology, and practice.
- Practice SQL programming through a variety of database problems.
- Demonstrate the use of concurrency and transactions in database.
- Design and build database applications for real world problems.

Content	Hours
Unit 1: Databases and Database Management Systems Introduction to databases – Characteristics of database approach, Users of the databases, Advantages of using the DBMS approach, Data Models, Schemas and instances; Three Schema Architecture and Data Independence; Database languages and interfaces, Database System Environment, Classification of Database Management Systems	12
Unit 2: Data Models and Database Design Classifications of Data Model, Data Modelling using the Entity Relationship model, Enhanced ER Data Model, Functional Dependencies, Relational Data model and Relational Database constraints Normalization, Object oriented relational databases	10
Unit 3: Query processing and Query optimization Query Processing – Basic steps in query processing – Measures of Query cost – Selection Operation – Basic Algorithms – Selection using indices – Evaluation of Expressions Materialization – Pipelining, Query Optimization – Transformation of Relational Expressions, Examples of Transformations Estimating Statistics of Expression Results.	10
Unit 4: Transaction and Recovery Transactions – Atomicity - Transaction State – Concurrent Execution – Serializability – Recoverability Implementation of Isolation – Testing for Serializability - Concurrency control; Types of concurrency protocols - Lock based protocols; Timestamp based protocols; Recovery systems – log-based recovery; Data Versioning	10
Unit 5: Database System Architectures Database System Architectures – Centralized and Client – Server Architectures – Server System Architectures - Parallel Databases – I/O parallelism, inter query and intra query parallelism, inter and intra operation parallelism, design of parallel systems. Homogeneous and Heterogeneous Databases – Distributed data storage, Distributed transactions; Commit protocols; concurrency control	10

References Books:

1. Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.
2. Database management systems, Ramakrishnan, and Gehrke, 3rd Edition, 2014, McGraw Hill.
3. Abraham Silberschatz, Henry F. Korth and S. Sudarshan's Database System Concepts 6th Edition Tata McGraw Hill Education Private Limited

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Identify, analyze and define database objects, enforce integrity constraints on a database using RDBMS	L2
CO2	Use Structured Query Language (SQL) for database manipulation and also demonstrate the basic of query evaluation.	L2
CO3	Design and build simple database systems and relate the concept of transaction, concurrency control and recovery in database	L3
CO4	Develop application to interact with databases, relational algebra expression.	L3
CO5	Develop applications using tuple and domain relation expression from queries.	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	x		x									
CO2		x			x							
CO3	x		x									
CO4	x			x								
CO5		x	x									

MCA105T: Data Structures and Algorithms

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Analyze step by step and develop algorithms to solve real world problems.
- Evaluate the Expressions like postfix, prefix conversions.
- Implementing various data structures viz. Stacks, Queues, Linked Lists, Trees and Graphs.
- Understanding various searching & sorting techniques.
- Be able to compare functions using asymptotic analysis and describe the relative merits of worst-, average-, and best-case analysis.

Content	Hours
Unit 1: Introduction and Linear Lists Introduction - Data Structure Definition, Classifications, Operations; Linear List Concepts. Linked List - Linked List Algorithms; Building a Linked List; Implementation; Linked List Abstract Data Type; Processing a Linked List; Linear List Applications; Complex Linked List Structures; Circular and Doubly Linked List; Applications	10
Unit 2: Stack and Queue Stack - Basic Stack Operations; Linked List implementation; Stack Applications; Conversion of Infix to Postfix; Evaluation of Postfix; Decimal to Binary conversion. Queue - Operations; Classifications; Linked List Implementation; Applications; Categorizing Data, Queue Simulations. Binary Tree: Concepts; Traversals; Binary Search Trees; AVL Trees; Multi-way Trees: B-Tree, Graph: Storage Structures; Traversal Algorithms;	12
Unit 3: Basic and Mathematical Concepts of Algorithm Fundamentals of Algorithmic Solving; Important Problem types; Asymptotic Notations and Basic Efficiency Classes. Mathematical Analysis of Non-recursive Algorithm; Mathematical Analysis of Recursive Algorithm; Example: Fibonacci Numbers; Empirical Analysis of Algorithms.	10
Unit 4: Algorithmic Design Techniques - I Divide and conquer - Merge sort; Quick Sort; Binary Search; Dynamic Programming - Warshall's and Floyd's Algorithm; Optimal Binary Search trees; Knapsack Problem and memory functions. Greedy Techniques - Prim's Algorithm; Kruskal's Algorithm; Dijkstra's Algorithm; Huffman trees.	10
Unit 5: Algorithmic Design Techniques - II Backtracking - n-Queen's Problem; Hamiltonian Circuit problem; Subset-Sum problem; Branch and bound - Assignment problem; Knapsack problem; Traveling salesman problem.	10

References Books:

1. Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures in C, 2nd Ed, Universities Press, 2014.
2. Seymour Lipschutz, Data Structures Schaum's Outlines, Revised 1st Ed, McGraw Hill, 2014.
3. Gilberg & Forouzan, Data Structures: A Pseudo-code approach with C, 2nd Ed, Cengage Learning, 2014.
4. Reema Thareja, Data Structures using C, 3rd Ed, Oxford press, 2012.
5. Jean-Paul Tremblay & Paul G. Sorenson, An Introduction to Data Structures with Applications, 2nd Ed, McGraw Hill, 2013.

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Explore different data structures, its operations..	L2
CO2	Demonstrate the concept of recursion and Queue.	L2
CO3	Apply the concept of Linked list, Trees and Graphs in problem solving	L3
CO4	Implement all data structures in a high-level language for problem solving	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	X											X
CO2					X							
CO3		X										X
CO4					X							

MCA106T: Theory of Computation

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Define and discuss the concept of formal grammar, formal language, regular expression and automata machine.
- Design finite automata and push down automata (PDA) machines for given formal languages or computational real-world problem statements.
- Understand the capability of Turing machine and design Turing Machine for context-sensitive languages or computational real-world problem statements.
- Choose and design appropriate automata for modeling the solution for various computational engineering problems.
- Understand the concepts of tractable & untractable problems and able to decide a given problem is tractable or not. Mapping of COs and POs.

Content	Hours
Unit 1: Review of Mathematical Terms and Theory Basic Mathematical Notations and Set Theory, Logic Functions and Relations, Language Definitions, Mathematical Inductions and Recursive Definitions. Finite Automata: Deterministic and Non Deterministic Finite Automata, U-Transitions, Conversion from NFA to DGA, Kleene's Theorem, Regular and Non Regular Languages.	12
Unit 2: Context Free Grammar Introduction to CFG, CFG and Known Languages, Unions, Concatenations and *'s Notations and CFL, Derivatives of Trees and Ambiguity and Unambiguous CFG and Algebraic Expressions, Normal Forms and Simplified Forms. Pushdown Automata, CFL and NFL: Introduction to PDA, Definition, DPDA, PDA Corresponding to CFG, CFG Corresponding to PDA, Introduction to CFL, Intersections and Complements of CFL, Decisions Problems and CFL.	10
Unit 3: Turing Machines, Recursive Language Model of Computation and Church Turning Thesis, Definitions of Turing Machine, TM and Language Acceptors, Variations of TM, Non Deterministic TM, Universal TM, Enumerable and Language, Recursive and Non Recursive Enumerable.	10
Unit 4: Computation Functions, Measuring, Classifications And Complexity Primitive Recursive Functions, Halting Problem, Recursive Predicates and Some Bounded Operations, Unbounded Minimizations and μ -Recursive Functions, Godel Numbering, Computable Functions and μ -Recursive, Numerical Functions.	10

Unit 5: Tractable and Intractable Problems

Growth Rate and Functions, Time and Speed Complexity, Complexity Classes, Tractable and Possibly Intractable Problems, P and Np Completeness, Reduction of Time, Cook's Theorem, Np-Complete Problems.

10

References Books:

1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, "Introduction to Automata Theory, Languages and Computation", 3rd Edition, Pearson Education, 2011.
2. John C Martin, "Introduction to Languages and Automata Theory", 3rd Edition, Tata McGraw-Hill, 2007.
3. Daniel I.A. Cohen, "Introduction to Computer Theory", 2nd Edition, John Wiley and Sons, 2009.
4. Thomas A. Sudkamp, "An Introduction to the Theory of Computer Science, Languages and Machines", 3rd Edition, Pearson Education, 2006.

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Will apply knowledge of computing and mathematics appropriate to the discipline	L2
CO2	Learn about Automata theory and its application in Language Design	L2
CO3	Learn about Turing Machines and Pushdown Automata and understand Linear Bound Automata and its applications	L3
CO4	Discuss key notions of computation, such as algorithm, computability, decidability, reducibility, and complexity, through problem solving	L3
CO5	Solve computational problems regarding their computability and complexity and prove the basic results of the theory of computation	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1			X									
CO2								X				
CO3			X							X		
CO4					X							
CO5	X											

MCA107P: Advanced Java Programming Lab

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

The main objective of this laboratory is to put into practice computational thinking. The students will be expected to write, compile, run and debug Python programs to demonstrate the usage of

- Demonstrate working of recursive functions using Java.
- Demonstrate working of constructors, inheritance and runtime polymorphism.
- Demonstrate working of Tree Set and Stack operations.
- Exception handling in Java.
- Demonstrate the usage of Swings.

Content
1. Write a program to convert a given Decimal number to Binary, Octal and Hexadecimal using recursive functions.
2. Write a program to explain the concept of constructor overloading.
3. Explain the concept of passing objects as parameters by adding two distances given in feet and inches.
4. Write a program to implement inheritance Concept in Java.

5. Write a program to explain the concept of runtime polymorphism in Java.
6. Write a program to implement producer consumer problem using thread concept.
7. Write a program to create object for Tree Set and Stack and use all methods.
8. Write a program to implement Exception handling in Java.
9. Write a program to implement the Concept of Interface in Java.
10. Write a program to get file name at runtime and display number of lines and words in that file.
11. Write a program to list files in the current working directory depending upon a given pattern.
12. Create a Frame for Student Registration containing all the fields Name, Age, Contact, Father's Name, Annual Income and a submit button. Perform field validations.
13. Write a program to demonstrate the usage of Swings.
14. Write a program to perform the following operation using JDBC: Insert, Update, Delete and Select Data.

MCA108P: Data Structures and Algorithms Lab

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

The main objective of this laboratory is to put into practice computational thinking. The students will be expected to write, compile, run and debug Python programs to demonstrate the usage of

- Demonstrate working of linked list.
- Demonstrate working of stack and queues.
- To find tree traversals, minimum cost spanning tree using prims and kruskals methods,
- N Queen's problem using Backtracking.
- 0/1 Knapsack problem using Dynamic Programming.

Content
1. Write a menu driven program to implement linear and binary search to find the location of the first occurrence of an item.
2. Write a menu driven program to create a singly linked list and to perform insert and delete operations.
3. Write a menu driven program to perform insert and delete operations in a circular linked list.
4. Write a menu driven program to perform operations on a stack (linked list implementation) (push, pop, and peek).
5. Write a program to convert the given infix expression into its postfix form.
6. Write a menu driven program to perform operations on a circular queue (enqueue, dequeue and peek). (linked list implementation)
7. Write a program to sort the array in ascending/descending order using a) Quick sort b) Merge sort
8. Write a program to find tree traversals of a binary tree.
9. Write a program to find Minimum cost spanning tree of a given undirected graph using Prim's algorithm.
10. Write a program to find Minimum cost spanning tree of a given undirected graph using Kruskal's algorithm.
11. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm
12. Implement N Queen's problem using Backtracking and prove its efficiency.
13. Find a subset of a given set $S = \{s_1, s_2, \dots, s_n\}$ of n positive integers whose sum is equal to a given positive integer d. A suitable message is to be displayed if the given problem instance doesn't have a solution and prove its efficiency
14. Implement 0/1 Knapsack problem using Dynamic Programming.

SECOND SEMESTER

MCA201T: Artificial Intelligence

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- To enable the students to understand the basic principles of Artificial Intelligence in various applications.
- Learn the methods of solving problems using Artificial Intelligence.
- Learn the knowledge representation techniques, reasoning techniques and planning.
- Introduce the concepts of Expert Systems and machine learning.
- Identify problems where artificial intelligence techniques are applicable.

Content	Hours
Unit 1: Introduction to Artificial Intelligence Introduction to AI, Definition, Foundations of AI, AI Technique -Tic-Tac-Toe. Production systems, Production system characteristics,8-puzzle problem. State Space Search : Uninformed search strategies – breadth first search, depth first search, Informed Search-Greedy Best First Search, A*Search, Hill Climbing, Simulated Annealing, Local beam Search, Mean end Analysis.	12
Unit 2: Intelligent Agents & Knowledge Representation Terminologies, Types of Agents, Properties of an Environment, Knowledge Representation, Propositional Calculus, Predicate Calculus, Rule based, Semantic Nets and Frames.	10
Unit 3: Artificial Intelligence Languages and Tools LISP-Environment, Program structure, Basic Syntax, Variables, Constants, Operators, decision, functions, lists; PROLOG-Syntax, clauses and predicates; Python libraries for AI-AIMA, pyDatalog, SimpleAI, EasyAI; MATLAB-getting started, Simple Programs; R-introduction to AI tools.	10
Unit 4: Applications of Artificial Intelligence Game Playing- Minimax Search Procedure, Alpha-Beta Cutoff, Watson and Jeopardy Challenge, Creating a Super Mario.	10
Unit 5: Expert System and Applications Architecture, Knowledge Representation- Frame based, Rule Based, Knowledge Acquisition, Learning in Expert Systems, Inference Process, Applications of Expert Systems, Examples- MYCIN, Dendral, SHINE, Expert System in Agriculture, Expert System in Environmental Management.	10

References Books:

- Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Mc Graw Hill 2008.
- Stuart Russel and Peter Norvig “AI – A Modern Approach”, 2nd Edition, Pearson Education 2007
- Peter Jackson, “Introduction to Expert Systems”, 3 rd Edition, Pearson Education, 2007.
- Dan W. Patterson, “Introduction to AI and ES”, Pearson Education, 2007. (Unit-III).
- Carl Townsend, “Introduction to Turbo PROLOG”, BPB Publication.
- Ivan Bratko,”Prolog Programming for Artificial Intelligence”, 3 rd Edition, Pearson Education

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

CO1	Gain a historical perspective of AI and its foundations.
CO2	Become familiar with basic principles and strategies of AI towards problem solving
CO3	Understand and apply approaches of inference, perception, knowledge representation, and learning.
CO4	Understand the various applications of A

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Identify problems that are amenable to solution by AI methods.	L2
CO2	Identify appropriate AI methods to solve a given problem.	L2
CO3	Formalize a given problem in the language/framework of different AI methods	L2
CO4	Implement basic AI algorithms for the given problem.	L3
CO5	Design and carry out an empirical evaluation of different algorithms on a problem formalisation, and state the conclusions that the evaluation supports.	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		x		x								
CO2	x		x									
CO3		x	x									
CO4	x		x									
CO5	x	x										

MCA202T: Web Technologies

Total Teaching Hours: 52

No. of Hours / Week: 04

Course learning objectives:

- To comprehend the basics of the internet and web terminologies.
- To introduce scripting language concepts for developing client-side applications.
- To practice server-side programming features – PHP, JSP.
- To be familiar with database applications
- To build dynamic web pages with validation using Java Script objects and by applying different event handling mechanisms.

Content	Hours
Unit-I: Client-side technologies The world wide web, HTML5: basic structuring tags, form, table, multimedia, CSS3: selectors, box model, styling on tables, divisions, alignments. Introduction to XML, Defining XML tags, their attributes and values, Document type definition, XML Schemas, DOM and DOM Events, XHTML, Introduction to AJAX	12
Unit-II: Introduction to JavaScript Syntax, variables, datatypes, operators, control structures, functions, control statements, arrays, form validation, JS objects – Date, Math, String, event handling, JS Browser Object Model – window object, history object, navigator object, screen object, HTML Document Object Model	10
Unit-III: Introduction to Angular JS Introduction to angular JS, features of angular JS, Architectural concepts, expressions, built-in filters, basic usage of filters, introduction to directives, controllers, and modules, working with angular forms and events, form validation.	10
Unit-IV: Introduction to PHP Introduction to PHP, PHP Arrays and Super globals, Arrays, Reading/Writing Files, Object-Oriented Overview, Error Handling and Validation,	10
Unit-V: Accessing Database Introduction to MongoDB: Setting up MongoDB, hosting and authenticating into database, Model Creation, Managing Database Connections, and Performing basic operations – Insert, update, delete and search	10

Reference

1. Randy Connolly, Ricardo Hoar, "Fundamentals of Web Development", 1st Edition, Pearson Education India.
2. Kirupa Chinnathambi, "JavaScript Absolute Beginner's Guide", Que Publishing, 1st Edition, 2017.

3. Robert W Sebesta, “Programming the World Wide Web”, Pearson, 7th Edition, 2013.
4. Mark Pilgrim, “HTML5 Up and Running”, O’Reilly, 1st Edition, 2015.
5. Learning PHP, MySQL and JavaScript By Robin Nixon –O’REILLY Publications

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Apply the features JQuery for the given web based problem	L2
CO2	Demonstrate the development of XHTML documents using JavaScript and CSS.	L2
CO3	Illustrate the use of CGI and Perl programs for different types of server side applications.	L3
CO4	Design and implement user interactive dynamic web based applications.	L3
CO5	Demonstrate applications of Angular JS and JQuery for the given problem.	L2
CO6	Apply the concept and usages web based programming techniques.	L2
CO7	Learning and Developing XHTML documents using JavaScript and CSS.	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	x			x								
CO2		x				x						
CO3	x		x									
CO4		x			x							
CO5	x		x									
CO6	x		x									
CO7		x				x						

MCA203T: Advanced Python Programming

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- Learn Syntax and Semantics and create Functions in Python.
- Handle Strings and Files in Python.
- Understand Lists, Dictionaries and Regular expressions in Python.
- Implement Object Oriented Programming concepts in Python.
- Build Web Services and introduction to Network and Database Programming in Python.

Content	Hours
Unit 1: Python Basics Introduction, Features, Identifiers; Keywords; Statements and Expressions; Variables; Operators; Precedence and Association; Data Types; Indentation; Comments; Built-in Functions- Console Input and Console Output, Type Conversions; Python Libraries; Importing Libraries with Examples. Control Flow Statements- if, else, elif, while loop, break, continue statements, for loop Statement; range () and exit () functions.	12
UNIT 2: Functions, Strings Fruitful functions: return values, parameters, local and global scope, function composition, User Defined Functions – Defining, Calling, Types of Functions, Arguments; Lambda Function; Importing Modules; recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays.	10
UNIT 3: Lists, Tuples, Dictionaries, Regular Expression Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing - list comprehension. Regular Expressions – Concept of regular expression, various types of regular expressions, using match function.	10
UNIT 4: Files, Modules, Packages, Database connectivity	10

Files and exception: text files, reading and writing files, format operator; command line arguments, errors and exceptions, handling exceptions, modules, packages. Database Programming: Connecting to a database, Creating Tables, INSERT, UPDATE, DELETE and READ operations, Transaction Control, Disconnecting from a database, Exception Handling in Databases.	
UNIT 5: Data Analysis and Data Visualisation: Data Analysis: NumPy- Introduction to NumPy, Array Creation using NumPy, Operations on Arrays; Pandas- Introduction to Pandas, Series and DataFrames, Creating DataFrames from Excel Sheet and .csv file, Dictionary and Tuples. Operations on DataFrames. Data Visualisation: Introduction to Data Visualisation; Matplotlib Library; Different Types of Charts using Pyplot- Line chart, Bar chart and Histogram and Pie chart.	10

References:

1. Think Python How to Think Like a Computer Scientist, Allen Downey et al., 2nd Edition, Green Tea Press. Freely available online @ <https://www.greenteapress.com/thinkpython/thinkCSpy.pdf>, 2015.
2. Introduction to Python Programming, Gowrishankar S et al., CRC Press, 2019.
3. Python Data Analytics: Data Analysis and Science Using Pandas, matplotlib, and the Python Programming Language, Fabio Nelli, Apress®, 2015
4. Advance Core Python Programming, MeenuKohli, BPB Publications, 2021.
5. Core PYTHON Applications Programming, Wesley J. Chun, 3rd Edition, Prentice Hall, 2012.
6. Automate the Boring Stuff, Al Sweigart, No Starch Press, Inc, 2015.
7. Data Structures and Program Design Using Python, D Malhotra et al., Mercury Learning and Information LLC, 2021.

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Understand and comprehend the basics of Python programming.	L2
CO2	Apply knowledge in real time applications	L3
CO3	Apply the Data Pre-processing & Data Wrapping	L3
CO4	Demonstrate the Web Scraping And Numerical Analysis	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2	X		X									X
CO3					X					X		
CO4		X										

MCA204T: Operating Systems and Linux

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objective:

- Describe the basics of the operating systems, mechanisms of OS to handle processes, threads, and their communication.
- Analyze the memory management and its allocation policies.
- Illustrate different conditions for deadlock and their possible solutions..
- Discuss the storage management policies with respect to different storage management technologies.
- To understand and make effective use of linux utilities and shell scripting language to solve problems.

Content	Hours
Unit 1: Operating System Overview Computer System Overview, Basic Elements, Instruction Execution, Interrupts Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. Operating System Overview, Objectives and functions, Evolution of Operating System, Computer System Organization, Operating System Structure and Operations, System Calls, System Programs, OS Generation and System Boot.	10
Unit 2: Process and Storage Management Processes, Process Concept, Process Scheduling, Operations on Processes, Interposes Communication; Threads, Overview, Multicore Programming, Multithreading Models; Process Synchronization, Critical Section Problem, Mutex Locks, Semaphores, Monitors; CPU Scheduling and Deadlocks.	10
Unit 3: Memory Management Main Memory, Contiguous Memory Allocation, Segmentation, Paging, 32 and 64 bit architecture Examples; Virtual Memory, Demand Paging, Page Replacement, Allocation, Thrashing; Allocating Kernel Memory, OS Examples.	10
Unit 4: File and I/O System Mass Storage Structure, Overview, Disk Scheduling and Management; File System Storage, File Concepts, Directory and Disk Structure, Sharing and Protection; File System Implementation, File System Structure, Directory Structure, Allocation Methods, Free Space Management, I/O Systems. Disk Scheduling.	10
Unit 5: Linux and Shell Programming Introduction, Features, Shell Programming, Types of Shells, Shell Syntax, Pipes and Redirection, Environment Variables, Working with Files. Kernel, Overview, Kernel Related Commands; Creation of Static & Dynamic Libraries, Module Programming, The Hello World Module, Module Stacking, Module Parameters, Registering a System Call, System Call Handler, Service routine.	12

Reference Books:

1. Abraham Silberschatz and Peter Baer Galvin, "Operating System Concepts", 7th Edition, Pearson Education, 2002.
2. H.M.Deitel, "Operating Systems", Pearson Learning Solutions, 3rd Edition, 2003.
3. William Stallings, "Operating Systems", 6th Edition, Pearson Education, 2010.
4. Stuart, "Operating systems: Principles, Design and Implementation", 1st Edition 2008, Cengage Learning India

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Analyse the basic Operating System Structure and concept of Process Management	L2
CO2	Analyse the given Synchronization/ Deadlock problem to solve and arrive at valid conclusions	L2
CO3	Analyse OS management techniques and identify the possible modifications for the given problem context	L2
CO4	Ability to design and solve synchronization problems.	L3
CO5	Ability to simulate and implement operating system concepts such as scheduling, Deadlock management, file management, and memory management.	L3

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	x				x							
CO2				x			x					
CO3	x		x									
CO4			x		x							
CO5		x		x								

MCA205T: Network and Information Security

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- To understand basics of Network and Information Security
- To understand basics of Cryptography and Network Security
- To be able to secure a message over insecure channel by various means
- To learn about how to maintain the Confidentiality, Integrity and Availability of a data
- To understand various protocols for network security to protect against the threats in the networks

Content	Hours
Unit 1: : Introduction Uses of computer networks; Network hardware; Network software; Reference models – OSI Reference models, Design issues of the OSI Reference model, TCP/IP Model A comparison of OSI TCP/IP reference models	10
Unit 2: Network Security Fundamentals Introduction; Computer Security Concepts; The OSI Security Architecture; Types of Security Attacks; Security Services; Security Mechanisms, A Model for Network Security; Standards; Symmetric Encryption and Message Confidentiality - Symmetric Encryption Principles; Symmetric Block Encryption Algorithms; Cipher Block Modes of Operation.	12
Unit 3: Public-Key Cryptography and Message Authentication Approaches to Message Authentication; Secure Hash Functions; Message Authentication Codes; Public-Key Cryptography Principles; Public-Key Cryptography Algorithms; RSA and Diffie-Hellman Key Exchange; Digital Signatures.	10
Unit 4: Key Distribution and User Authentication Symmetric Key Distribution using Symmetric Encryption; Kerberos; Key Distribution Using Asymmetric Encryption; X.509 Certificates; Public-Key Infrastructure; Federated Identity Management; Transport-Level Security - Web Security Considerations; Secure Socket Layer and Transport Layer Security; Secured TCP extension of three-way handshaking method; HTTPS.	10
Unit 5: Wireless Network Security and E-Mail IEEE 802.11; Wireless LAN Overview; IEEE 802.11i; Wireless LAN Security; Wireless Application Protocol Overview; Wireless Transport Layer Security; WAP End-to-End Security; Electronic Mail Security - Pretty Good Privacy, S/MIME, Domain Keys Identified Mail. IP Security - IP Security Overview, IP Security Policy; Intruders, Intrusion detection	10

Reference Books:

1. Michael E. Whitman and Herbert J. Mattord: “Principles of Information Security”, 4th Edition, Thomson, 2012.
2. William Stallings: “Network Security Essentials Applications and Standards”, 4th edition, Person Education, 2012.
3. Behrouz a Forouzan, Debdeep Mukhopadhyay: “Cryptography and Network Security”, 2nd edition, Tata McGraw-Hill, 2011.
4. Deven N. Shah: Mark Stapms Information Security Principles & Practice, 1st edition, Wiley India, 2010.

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Apply the basic concepts of networks like protocol, internet and OSI layers	L3
CO2	Analyze the working of Physical Layer.	L3
CO3	Demonstrate the various Switching networks	L3
CO4	Analyze the Data Link Layer	L3

Mapping of COs and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	x				x							
CO2		x										
CO3	x				x							
CO4		x										

MCA206T: Cloud Computing

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

- To provide knowledge of cloud architecture, deployment models.
- To introduce broad perceptive of cloud services.
- To introduce about storage and database management in cloud computing.
- To make them understand about resource management in cloud computing
- To make them familiar with the various cloud security issues and research trends in cloud

Content	Hours
Unit 1: Introduction Cloud Computing at a glance – Historical Developments- Building Cloud Computing Environments- Computing Platforms and Technologies – Virtualization- Characteristics of Virtualized Environments – Taxonomy of Virtualization – Virtualization and Cloud Computing – Pros and Cons of Virtualization	12
Unit 2: Cloud Architecture and Models Introduction – Cloud Reference Model, Architecture; Types of Clouds – Economics of the Cloud – Open Challenges – Cloud Application Platform – Cloud deployment model: Public clouds – Private clouds – Community clouds – Hybrid clouds – Advantages of Cloud computing	10
Unit 3: Cloud Solutions Global cloud infrastructure-Storage –Compute- Database – Networking – Implementing Elasticity, High Availability, and Monitoring – Building Microservices and Serverless Architectures	10
Unit 4: Cloud Security Shared responsibility model in cloud –security design principles-securing Access to Cloud Resources-securing access to infrastructure- Protecting data- Logging and monitoring	10
Unit 5: Case Study Networking and content delivery, Auto scaling, Load balancing ,Storage , database services in AWS	10

Reference

1. Cloud Computing – Insight into New Era Infrastructure, Dr. Kumar Saurabh, Wiley India, 2011.
2. Cloud Computing, Roger Jennings, Wiley India, 2009.
3. Cloud Computing Explained, John Rhoton, Recursive Press,2009.
4. Cloud Computing Bible, Barry Sosinsky, Wiley, 2011.

Course outcome (Course Skill Set)

At the end of the course the student will be able to :

Sl. No.	Description	Blooms Level
CO1	Demonstrate the fundamental and core concepts of cloud computing	L2
CO2	Compare between parallel and distributed computing	L3
CO3	Investigate the system virtualization and outline its role in enabling the cloud computing system model	L2
CO4	Compare different deployment and service models of cloud to develop different variety of applications	L2

Mapping of COS and POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	X											
CO2	X	X			X					X		
CO3	X	X										
CO4	X	X	X									

MCA207P: Web Technology Lab

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

The main objective of this laboratory is to put into practice computational thinking. The students will be expected to write, compile, run and debug Python programs to demonstrate the usage of

- HTML forms and style sheets using CSS.
- Create an application using JavaScript.
- Develop a PHP program with SQL connection.
- AJAX and AngularJS implementation.
- Create a XML program.

Content
1. Develop and demonstrate the usage of inline and external style sheet using CSS.
2. Create a table of a minimum size of 4 rows and 4 columns (including headers) using HTML. Add styles to borders, and cells using CSS. Add a button on the page to hide and show the table.
3. Create an application with different inputs such as radio buttons, date picker, numeric, etc., and display the entered values right below the form.
4. Develop a HTML Form, which accepts any Mathematical expression. Write JavaScript code to Evaluates the expression and Displays the result.
5. Create an employee registration form and validate the fields using JavaScript: date, confirm password, password strength, email, number and display custom error messages and success indicators next to each field.
6. Create a form for Student information. Write JavaScript code to find Total, Average, Result and Grade.
7. Create a form for Employee information. Write JavaScript code to find DA, HRA, PF, TAX, Gross pay, Deduction and Net pay.
8. Develop a to-do list application that allows users to add, remove, and reorder tasks using drag-and-drop. Implement event handling for user interactions and validation to prevent duplicate tasks.
9. Create an AngularJS app where users can input two numbers and select an operation to perform. The result should be displayed dynamically.
10. Develop an AngularJS app that displays a dropdown list of countries. When a country is selected, another dropdown list of cities (specific to the selected country) is populated
11. Write an AJAX implementation where the content of a text file will be displayed on some part of the page without altering the existing text or refreshing the page. Instead, use a button to update the contents.
12. Write a PHP script to create a file and perform both read and write operations.
13. Write a PHP program to read student data from an XML file and store into the database. Retrieve and display.
14. Develop a PHP program to design student information form using database.

MCA208P: Advanced Python Programming Lab

Total Teaching Hours: 52

No. of Hours / Week: 04

Course Learning objectives:

The main objective of this laboratory is to put into practice computational thinking. The students will be expected to write, compile, run and debug Python programs to demonstrate the usage of

- Demonstrate working of control structures.
- Demonstrate working with lists, tuples and dictionaries in Python.
- Data Manipulation operations using arrays.
- Data Manipulation operations using strings.
- Usage of file operations.

Content
1. Write a python program to demonstrate the usage of bitwise operators.
2. Write a python program to find largest and smallest of three numbers.
3. Write a python program to print 'n' terms of Fibonacci series of numbers
4. Write a python program to find factorial of a number using Recursion.
5. Write a Python program to find number of occurrences of each vowel present in the given string.
6. Write a program to create, concatenate and print a string and accessing sub-string from a given string.
7. Write a Python program to find whether the given pattern is available in the given string or not?
8. Write a python program to check whether the given mail id is valid gmail id or not?
9. Write a program to create a Lambda Function to find biggest of given values.
10. Write a program to create, append, and remove lists in python.
11. Write a program to demonstrate working with tuples in python.
12. Write a program to demonstrate working with dictionaries in python.
13. Write a python program to perform read and write operations on a file.
14. Write a python program to copy the contents of one file to another file.
